

Assignment 3 - bonus point

Min Zhu

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Exercise 5.1

To improve the model performance, the following strategies were adopted:

1. Widening the network
2. Data augmentation
3. Optimizer switch
4. Fine-tuning

As a result, the final model achieves an accuracy of **70.57%**.

Widening the network

To increase model capacity, two strategies are adopted: widening each layer and increasing the number of filters. Specifically, the configuration is set to **f=4**, **nf=40**, **nh=300**.

Under this setting, the model achieves a test accuracy of **65.70%**, which serves as the baseline model for subsequent improvements.

Data augmentation

Initially, I experimented with random translation within the range $x, y \in [-3, 3]$, as implemented in Assignment 2. However, even with pretraining, this approach significantly increased the effective dataset size. Combined with the relatively large network architecture used in this setting, the computational cost of training became extremely high.

As an alternative, I adopted lighter data augmentation techniques: random crop + horizontal flip. These methods introduce variability with much lower computational overhead and lead to a slight improvement in performance, increasing the test accuracy to 65.76%.

In addition, I experimented with the Cutout technique under various parameter settings. However, this resulted in degraded model performance, and therefore it was not retained in the final configuration.

Table 1: Data Augmentation

Method	Aug Parameters	Final Test Acc	Improvement
random crop + horizontal flip	flip_prob=0.5, pad=4	65.76%	0.06%
crop, flip, cutout	cut_p=0.5, size=12, fill=0.0	64.05%	-1.71%
crop, flip, cutout	cut_p=0.2, size=8, fill=0.0	63.20%	-2.56%

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Final accuracy summary:  
f=4, nf=40, nh=300: final test acc=0.657600, train time=560.32s
```

Figure 1: Data augmentation

Optimizer switch

The current model is trained using standard SGD. To enhance optimization efficiency, SGD with momentum is adopted. With an initial setting of $\eta_{max} : 5e - 2, \beta(\text{momentum}) = 0.8, \lambda = 2.5e - 3$, the model shows a marginal improvement, achieving a test accuracy of 65.77%. This suggests that momentum contributes to more stable and efficient convergence, although the overall gain remains limited under the current hyperparameter setting.

Fine-tuning

Despite applying multiple improvement strategies, the overall performance gain remains limited, which may be attributed to suboptimal hyperparameter settings. Therefore, I proceed to perform hyperparameter tuning.

In this stage, I focus on adjusting the learning rate η_{max} and the regularization coefficient λ , while keeping the overall network architecture fixed. To reduce computational cost, each trial is trained for only 35% of the full training epochs during the tuning phase. The detailed experimental procedure is as follows:

1. First, keep λ fixed and change the value of η_{max} .

Table 2: Fine tuning1

η_{max}	λ	final test acc	train time (s)
0.04	0.0025	0.618000	241.15
0.05	0.0025	0.615000	210.73
0.06	0.0025	0.609100	202.72
0.07	0.0025	0.604600	206.91

2. Then, adopt the best $\eta_{max} = 0.04$,

Table 3: Fine tuning2

η_{max}	λ	final test acc	train time (s)
0.04	0.002	0.623900	189.75
0.04	0.0025	0.618000	199.59
0.04	0.003	0.607800	205.30

When $\eta_{max} = 0.04$ and $\lambda = 0.02$, the model yields the best performance. However, it is observed that both optimal values lie at the lower boundary of the explored parameter space. This suggests that the search range may not yet be sufficiently wide. Therefore, the range is further extended towards smaller values for both parameters to enable a more comprehensive hyperparameter search and potential further improvement.

3. Fix $\lambda = 0.002$, and explore when η_{max} is smaller:

Table 4: Fine tuning3

eta_max	lam	final test acc	train time (s)
0.02	0.002	0.606300	230.16
0.03	0.002	0.620600	227.75
0.035	0.002	0.623000	191.60
0.04	0.002	0.623900	195.61

This result shows that $\eta_{max} = 0.04$ yields the best.

4. Set $\eta_{max} = 0.04$, and experiment on smaller λ :

Table 5: Fine tuning4

η_{max}	λ	final test acc	train time (s)
0.04	0.0001	0.641400	410.36
0.04	0.0003	0.646100	402.13
0.04	0.0005	0.636800	414.08
0.04	0.0007	0.636400	403.43
0.04	0.001	0.638400	425.79
0.04	0.0015	0.629100	208.66
0.04	0.002	0.623900	170.84

After conducting a more fine-grained search over a smaller range, the setting $\{\eta_{max} = 0.04, \lambda = 0.0003\}$ achieves the best performance under 35% of the total training epochs. Based on this result, the model is further trained using the full number of epochs under the same hyperparameter setting to obtain the final performance evaluation.

Final Model

The best test accuracy achieved by the three-layer convolutional network is **70.58%**. The corresponding configuration is:

- **Learning rate (cyclical):** $\eta_{min} = 1 \times 10^{-5}$, $\eta_{max} = 4 \times 10^{-2}$
- **Regularization coefficient:** $\lambda = 3 \times 10^{-4}$
- **Momentum:** 0.8
- **Data augmentation:** $flip_{prob} = 0.5, pad = 4$
- **Batch size:** $n_{batch} = 100$
- **Training schedule (cycle lengths):** $n_s = [800, 1600, 3200]$
- **Network architecture:** $f = 4, n_f = 40, n_h = 300$

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Final accuracy summary:
f=4, nf=40, nh=300 | eta_max=0.04 lam=0.0003: final test acc=0.705700, train time=1072.65s
```

Figure 2: Best performance